

**Type: UGA**

**Maximum Marks: 120**

**Duration: 120 Minutes**

**Each correct answer carries 4 marks, unattempted question carries 1 mark, wrong answer carries 0 mark**

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**Q. 1** Let  $A$  be a symmetric idempotent matrix. Which of the following statements is correct?

- (a)  $\det(A) = 0$
- (b)  $\text{tr}(A)$  is equal to the number of columns of  $A$
- (c) All the diagonal elements of  $A$  lie between 0 and 1
- (d)  $\det(A) = \text{tr}(A)$

**Q. 2** The limit

$$\lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{1}{\binom{n}{k}}$$

equals

- (a) 0
- (b) 1
- (c) 2
- (d) the limit does not exist

**Q. 3** Let  $z_1, z_2, z_3$  be distinct complex numbers such that  $|z_1| = |z_2| = |z_3| > 0$ . If  $z_1 + z_2z_3, z_2 + z_3z_1$ , and  $z_3 + z_1z_2$  all are purely real, then  $z_1z_2z_3 =$

- (a) +1
- (b) -1
- (c) 2
- (d) cannot be found from the given information

**Q. 4** The remainder obtained when  $\binom{100}{50}$  is divided by 7 is

- (a) 0
- (b) 2
- (c) 3
- (d) 4

**Q. 5** Let  $a > 0, b > 0, c > 0, d > 0$  be real numbers such that

$$\sqrt{a^2 + b^2 - ab\sqrt{2}} + \sqrt{b^2 + c^2 - bc\sqrt{2}} + \sqrt{c^2 + d^2 - cd\sqrt{2}} = \sqrt{a^2 + d^2 + ad\sqrt{2}}.$$

Then,  $\frac{(a+c)(b+d)}{ad} =$ .

- (a) 2
- (b) 5
- (c)  $\sqrt{2}$
- (d) 3

- Q. 6** Let  $a, b, c$  be the lengths of the sides of a triangle. Suppose that  $p = \frac{a}{b} + \frac{b}{c} + \frac{c}{a}$  and  $q = \frac{a}{c} + \frac{c}{b} + \frac{b}{a}$ . Then  $|p - q|$  will always be
- less than or equal to 1
  - equals to 0
  - strictly greater than 1
  - strictly less than 1
- Q. 7** Suppose that for some primes  $p, q$ , the quantities  $p^2 + q^3$  and  $p^3 + q^2$  both are primes. Then the value of  $p + q + p^2 + q^2 + p^3 + q^3$  is
- 53
  - 52
  - 51
  - there can be infinitely many values.
- Q. 8** Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a continuous function such that  $f(x)f(f(x)) = 1$  for every real  $x$ . If  $f(1000) = 999$ , then the value of  $f(500)$  is.
- 1
  - $\frac{1}{999}$
  - $\frac{1}{500}$
  - none of the above
- Q. 9** Let  $x = \log_{15} 2$ ,  $y = \log_{10} 3$ ,  $z = \log_6 5$ . Then
- $xy + yz + zx + 2xyz > 1$
  - $xy + yz + zx + 2xyz = 1$
  - $xy + yz + zx + 2xyz < 1$
  - $xy + yz + zx + xyz = 1$
- Q. 10** Let  $A$  denote the subset of the set  $\{1, 11, 22, 31, \dots, 541, 551\}$  having the property that no two elements of  $A$  adds up to give 552. Then the maximum number of elements that  $A$  can have is
- 29
  - 27
  - 30
  - 28
- Q. 11** The value of the definite integral
- $$\int_0^4 \frac{\ln x}{\sqrt{4x - x^2}} dx$$
- is
- 1
  - $\ln 2$
  - $\ln 3$
  - 0

**Q. 12** Which of the following is/are always a factor of  $P(x) = (1+x+x^2+x^3+x^4+x^5+x^6+x^7)^2 - x^7$ ?

- (a)  $1 + x + x^2 + x^3 + x^4 + x^5 + x^6$
- (b)  $1 + x + x^2 + x^3 + x^4 + x^5 + x^6 + x^7 + x^8$
- (c)  $1 + x + x^2 + x^3 + x^4 + x^5 + x^6 + x^7$
- (d) both (a) and (b)

**Q. 13** Suppose that a point  $P$  moves in a plane in such a manner that there exist two fixed point  $A, B$  on the same plane so that the quantity  $PA^2 - PB^2$  always remains constant for every position of  $P$ . Then

- (a) The locus of  $P$  is a hyperbola with foci at  $A$  and  $B$
- (b) The locus of  $P$  is an ellipse with foci at  $A$  and  $B$
- (c) The locus of  $P$  is a straight line perpendicular to the line joining  $A$  and  $B$
- (d) The locus of  $P$  is a circle with  $AB$  as a diameter

**Q. 14** Let  $n > 1$  be a natural number, then the value of the expression

$$\frac{\sqrt{n^2 - 1} + \sqrt{n^2 - 2^2} + \cdots + \sqrt{n^2 - (n-1)^2}}{0.79n^2}$$

- (a) is always less than 1
- (b) initially is less than 1, but eventually becomes larger than 1
- (c) oscillates
- (d) none of the above

**Q. 15** The maximum value of the function

$$f(x) = \sqrt{4x - x^2 + 12} - \sqrt{10x - x^2 - 24}$$

is

- (a) 2
- (b)  $2\sqrt{3}$
- (c)  $\sqrt{3}$
- (d) 1

**Q. 16** Suppose that in  $\triangle ABC$ ,  $\angle ABC = 90^\circ$ . Points  $D, E$  are on the hypotenuse  $AC$  such that  $AD = DE = EC$ . If  $BD = 3$ ,  $BE = 4$ , then  $AC =$

- (a) 5
- (b) 2
- (c)  $3\sqrt{5}$
- (d) 7

**Q. 17** Number of real solution(s) to the equation

$$x + x^3 + \sin(x) = x^2 + x^6 + \sin(x^6)$$

is

- (a) 0
- (b) 1
- (c) 2
- (d) more than 2

**Q. 18** The number of prime(s) that can be expressed as  $[\frac{n^2}{3}]$ , (where  $n \in \mathbb{N}$ , and  $[\cdot]$  denotes the greatest integer function) is

- (a) 1
- (b) 2
- (c) 3
- (d) infinitely many

**Q. 19** Given that  $3 \sin \alpha + 4 \cos \alpha - 7 \cos \beta = 12$ . The value of  $3 \sin \beta - 4 \cos \beta + 25 \cos \alpha$  is

- (a) 12
- (b) 13
- (c) 24
- (d) cannot be obtained

**Q. 20** Let  $I_1 = \int_0^1 (1 - x^{2025})^{\frac{1}{2026}} dx$ , and  $I_2 = \int_0^1 (1 - x^{2026})^{\frac{1}{2025}} dx$ , then

- (a)  $I_1 > I_2$
- (b)  $I_2 > I_1$
- (c)  $I_1 = I_2$
- (d)  $I_1 + I_2 = 1$

**Q. 21** The lengths of the consecutive sides of a convex quadrilateral are  $a, b, c, d$ , and its area is  $\frac{ab+cd}{2}$ . Then which of the following is always true?

- (a) The quadrilateral must be a rectangle
- (b) The quadrilateral must be a trapezium
- (c) The quadrilateral must be a square
- (d) The quadrilateral must be cyclic

**Q. 22** Let  $X$  be a set with  $n$  elements. A function  $f : X \rightarrow X$  is called an idempotent function if  $f(f(x)) = f(x)$  holds for all  $x \in X$ . Let  $i(n)$  denote the number of idempotent function on  $X$ . Then

- (a)  $i(n) = \sum_{k=1}^n \binom{n}{k} k^{n-k}$
- (b)  $i(n) = \sum_{k=1}^n \binom{n}{k} k^{n+k}$
- (c)  $i(n) = \sum_{k=1}^{n-1} \binom{n}{k} k^{n-k}$
- (d)  $i(n) = \sum_{k=1}^n \binom{n}{k+1} k^{n-k}$

**Q. 23** Let  $f : [0, 1] \rightarrow \mathbb{R}$  be continuous, with

$$\begin{aligned}\int_0^1 f^2(x)dx &= \frac{e^2}{2} + \frac{11}{6}, \\ \int_0^1 e^x f(x)dx &= \frac{e^2}{2} + \frac{1}{2}, \\ \int_0^1 xf(x)dx &= \frac{4}{3}.\end{aligned}$$

Then

- (a)  $f(0) = 0$
- (b)  $f(0) = 1$
- (c)  $f(0) = 2$
- (d)  $f(0) = e$

**Q. 24** A bag contains  $n$  white and  $n$  red balls. Pairs of balls are drawn without replacement until the bag is empty. The probability that every pair consists of one white and one red ball, is

- (a)  $\frac{2^n}{\binom{2n}{n}}$
- (b)  $\frac{n^2}{\binom{2n}{n}}$
- (c)  $\frac{2^n}{\binom{n+1}{n}}$
- (d)  $\frac{n^n}{\binom{2n}{n}}$

**Q. 25** Given that

$$\frac{x^2}{x+1} + \frac{y^2}{y+2} + \frac{z^2}{z+3} = x + y + z + 106.$$

Then the value of the following expression

$$\frac{1}{4x+4} + \frac{1}{y+2} + \frac{9}{4z+12}.$$

is

- (a) 25
- (b) 12
- (c) 53
- (d) 28

**Q. 26** The number of continuous function(s)  $f : \mathbb{R} \rightarrow \mathbb{R}$  such that  $f(f(x)) + f(x) + x = 0$  holds for every real  $x$  is

- (a) infinitely many
- (b) 1
- (c) 0
- (d) cannot be evaluated from the given information

**Q. 27** A matrix  $O$  is called orthogonal if  $OO^\top = O^\top O = I$ , where  $O^\top$  denotes the transpose of  $O$ . Let  $A$  and  $B$  be two orthogonal square matrices with  $\det(A) + \det(B) = 0$ . Then

- (a)  $\det(A) = \det(B) = 0$
- (b)  $\det(A + B) = 0$
- (c)  $\det(A + B) = 1$
- (d)  $\det(A) > \det(B)$

**Q. 28** Let  $z_1, z_2$  and  $z_3$  denote the vertices of an equilateral triangle in the complex plane, and let  $z_0$  be the centroid of the triangle. Then  $\frac{z_1^2 + z_2^2 + z_3^2}{z_0^2} =$ .

- (a) 9
- (b) 4
- (c) 1
- (d) 3

**Q. 29** The value of the expression

$$\int_0^1 \frac{x^{2027} - x^3}{\ln x} dx$$

is

- (a)  $\ln 2027 - \ln 3$
- (b)  $\ln 507$
- (c)  $\ln 508$
- (d)  $\ln 1013$

**Q. 30** Let  $f$  be differentiable of  $[0, \infty)$ . Suppose that, for all  $x \geq 0$  we have  $|f(x)| \leq 2$ , and  $f(x)f'(x) \geq \cos x$ , Then

- (a)  $\lim_{x \rightarrow \infty} f(x) = 0$
- (b)  $\lim_{x \rightarrow \infty} f(x) = 1$
- (c)  $\lim_{x \rightarrow \infty} f(x) = \cos 1$
- (d)  $\lim_{x \rightarrow \infty} f(x)$  does not exist

**-End of the Question Paper-**